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**A Unified Forensic Framework for the
Detection of Multimodal Synthetic Media:
Identifying Artifacts in Deepfakes and
Generative AI Content**

A thesis submitted in partial fulfillment of the requirements
for the degree of
Bachelor of Science in Computer Science & Engineering

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Declaration

We hereby declare that this thesis, entitled “A Unified Forensic Framework for the Detection of Multimodal Synthetic Media: Identifying Artifacts in Deepfakes and Generative AI Content,” is our own original work carried out under the supervision of Mohammad Mahmudul Hasan, Department of Computer Science, American International University–Bangladesh. It has not been submitted, in whole or in part, to any other university or institution for the award of any degree or diploma. All sources of information and material used in this work have been duly acknowledged and cited.

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Approval

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Abstract

The rapid advancement of generative artificial intelligence has made it possible to synthesize convincing fake video and cloned speech independently, producing multimodal forgeries in which either stream—or both—may be manipulated. Existing audio-visual deepfake detectors typically collapse this rich problem into a single binary “real versus fake” decision, discarding forensically critical information about *which* modality was manipulated and *when*. This thesis presents a unified forensic framework that reformulates multimodal synthetic-media detection as a fine-grained attribution problem: given a video clip, the framework independently determines whether the visual stream and the acoustic stream are authentic or AI-generated, assigns the clip to one of four authenticity quadrants (real/real, real-video–fake-audio, fake-video–real-audio, fake/fake), and temporally localizes the manipulated segments in each stream. At the core of the framework is **DAVID-Net**, a disentangled dual-branch cross-modal transformer that separates modality-specific authenticity evidence from cross-modal consistency evidence, allowing it to detect internally consistent double-fakes that defeat synchronization-based detectors. To achieve generalization to unseen generators, we introduce **Quadrant-Aware Contrastive Pretraining (QACP)**, a self-supervised stage that constructs all four authenticity quadrants from pristine data alone—via self-blended face video, vocoder copy-synthesis audio, and cross-clip mismatch—and structures the embedding space with a factorized supervised-contrastive objective, without using any deepfake generator outputs. The framework is evaluated on FakeAVCeleb, AV-Deepfake1M, and LAV-DF, with cross-dataset validation on DFDC and KoDF, using per-modality AUC, four-class attribution accuracy, temporal localization AP, calibration, robustness, and fairness metrics. [TODO: Summarize final results here once experiments complete.] The complete system is deployed as a public web service comprising a FastAPI inference API and a Next.js user interface, enabling end users to verify whether the video and voice of a clip are genuine or AI-generated.

Keywords: deepfake detection, multimodal forensics, audio-visual learning, synthetic media, contrastive pretraining, temporal localization, generative AI.

Contents

Declaration	i
Approval	ii
Acknowledgment	iv
Abstract	v
Contents	vi
List of Figures	ix
List of Tables	x
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Research Questions	2
1.4 Objectives	2
1.5 Contributions	3
1.6 Scope and Limitations	3
1.7 Thesis Organization	3
2 Literature Review	4
2.1 Unimodal Video Deepfake Detection	4
2.2 Unimodal Audio Spoofing Detection	4
2.3 Audio-Visual Binary Detection	4
2.4 Audio-Visual Consistency and Synchronization	5

2.5	Temporal Localization of Partial Forgeries	5
2.6	Proxy-Forgery and Generator-Free Training	5
2.7	Summary of the Research Gap	5
3	Proposed Methodology	7
3.1	Problem Formulation	7
3.2	DAVID-Net Architecture	7
3.2.1	Design Principles	7
3.2.2	Encoders	8
3.2.3	Synchronization-Consistency Module	8
3.2.4	Disentangled Cross-Modal Fusion	8
3.2.5	Multi-Task Heads and Objective	8
3.3	Quadrant-Aware Contrastive Pretraining (QACP)	9
3.3.1	Motivation	9
3.3.2	Synthetic Quadrant Construction	9
3.3.3	Factorized Supervised-Contrastive Objective	9
3.4	System Architecture and Deployment	10
4	Experimental Setup	11
4.1	Datasets	11
4.2	Preprocessing	11
4.3	Implementation Details	12
4.4	Evaluation Metrics	12
4.5	Baselines	12
5	Results and Discussion	13
5.1	In-Domain Detection and Attribution	13
5.2	Cross-Dataset Generalization	13
5.3	Temporal Localization	14
5.4	Ablation Study	14
5.5	Calibration, Robustness, and Fairness	14
5.6	Qualitative Analysis	14
5.7	Efficiency and Deployment	14

5.8	Discussion	15
6	Conclusion and Future Work	16
6.1	Conclusion	16
6.2	Limitations	16
6.3	Future Work	16
	Bibliography	18
A	Supplementary Material	21
A.1	Repository and Reproducibility	21
A.2	Unified Manifest Schema	21
A.3	API Response Format	21
A.4	Additional Results	22

List of Figures

List of Tables

1.1	The four authenticity quadrants of a multimodal clip.	1
2.1	Capability comparison of prior work and the proposed framework. . .	6
3.1	QACP pseudo-classes constructed from pristine clips.	9
4.1	Datasets used in this study.	11
5.1	Per-modality and quadrant results on FakeAVCeleb (subject-disjoint test split). [TODO: fill]	13
5.2	Cross-dataset AUC: train on FakeAVCeleb + AV-Deepfake1M, test un- seen. [TODO: fill]	13
5.3	Ablations. Each row reports in-domain and cross-dataset AUC. [TODO: fill]	14

Chapter 1

Introduction

1.1 Background

Generative artificial intelligence can now synthesize photorealistic talking-head video and natural-sounding cloned speech with minimal effort and data. Face-swapping, facial reenactment, lip-synchronization models, text-to-speech (TTS), and voice conversion (VC) systems are publicly available, and their outputs are increasingly difficult for humans to distinguish from genuine recordings. Because the visual and acoustic streams of a clip can be manipulated *independently*, a single video can fall into one of four authenticity quadrants:

Table 1.1: The four authenticity quadrants of a multimodal clip.

Quadrant	Video	Audio	Typical origin
RVRA	Real	Real	Genuine recording
RVFA	Real	Fake	Voice clone / TTS dubbed over real footage
FVRA	Fake	Real	Face swap / reenactment over real speech
FVFA	Fake	Fake	Fully synthetic avatar (lip-sync + TTS)

1.2 Problem Statement

The overwhelming majority of published audio-visual deepfake detectors reduce this problem to a single binary decision: the clip is either “real” or “fake.” This formulation is forensically inadequate for three reasons. First, it provides no *attribution*: a journalist, court, or moderation system needs to know whether the face was swapped, the voice was cloned, or both. Second, binary detectors tend to shortcut on whichever modality is easier to classify, degrading accuracy on the harder stream. Third, modern

manipulations are increasingly *partial*: only a few seconds of audio or a short facial segment are altered, which a clip-level binary label cannot express.

1.3 Research Questions

This thesis addresses the following research questions:

- RQ1.** Can a single model produce independent, well-calibrated authenticity decisions for the visual and acoustic streams while remaining competitive with specialized unimodal detectors?
- RQ2.** Does explicit modeling of audio-visual synchronization improve detection of manipulations that break cross-modal coherence?
- RQ3.** Does disentangling modality-specific authenticity evidence from cross-modal consistency evidence improve generalization to unseen generators and datasets?
- RQ4.** Can the model temporally localize which segments of each stream were manipulated, and does this objective aid the clip-level decision?
- RQ5.** Is the detector robust to real-world degradations and fair across demographic subgroups?
- RQ6.** Can an embedding space pretrained on synthetically constructed quadrants built from pristine data alone generalize to real forgeries from unseen generators better than supervised training on generator outputs?

1.4 Objectives

- To design a unified forensic framework that detects, per modality, whether the video and the audio of a clip are real or AI-generated.
- To propose **DAVID-Net**, a disentangled cross-modal transformer with independent per-modality decision heads, a synchronization-consistency module, and per-modality temporal localization.
- To propose **Quadrant-Aware Contrastive Pretraining (QACP)**, a generator-free pretraining strategy for cross-generator generalization.
- To evaluate the framework rigorously across multiple benchmarks with cross-dataset, robustness, calibration, and fairness protocols.

- To deploy the detector as a public web application (FastAPI inference API on Hugging Face and a Next.js user interface).

1.5 Contributions

The main contributions of this thesis are:

1. **QACP**: a self-supervised pretraining stage that constructs all four authenticity quadrants from pristine data only and structures three embedding spaces with a factorized supervised-contrastive objective, including an explicit *real-but-mismatched* class that teaches the model that desynchronization is not forgery.
2. A reformulation of audio-visual deepfake detection as a multi-task, per-modality attribution and localization problem.
3. **DAVID-Net**, a disentangled dual-branch cross-modal transformer architecture implementing this reformulation.
4. A generalization-first evaluation protocol with leave-one-generator-out and cross-dataset testing as first-class results.
5. A complete, reproducible, publicly deployed detection system.

1.6 Scope and Limitations

This work targets talking-head style video with a single primary subject and speech audio, the dominant setting of current deepfake threats and benchmarks. Detection outputs are probabilistic assessments, not legal proof of authenticity. [TODO: Refine scope after experiments.]

1.7 Thesis Organization

Chapter 2 reviews related work in unimodal and multimodal deepfake detection. Chapter 3 presents the proposed framework, including DAVID-Net and QACP. Chapter 4 describes the datasets, preprocessing pipeline, implementation, and evaluation protocol. Chapter 5 reports experimental results and ablation studies. Chapter 6 concludes and outlines future work.

Chapter 2

Literature Review

This chapter surveys the five research threads most relevant to multimodal synthetic-media forensics and identifies the gap this thesis addresses.

2.1 Unimodal Video Deepfake Detection

Early detectors treated deepfake detection as frame-level image classification. XceptionNet trained on FaceForensics++ [1] became the canonical baseline, followed by EfficientNet ensembles that won the Deepfake Detection Challenge [2]. Frequency-domain methods exploit GAN upsampling fingerprints, while temporal methods such as LipForensics [3] and RealForensics [4] model mouth-motion dynamics for stronger cross-manipulation generalization. *Limitation:* these methods are blind to audio manipulation and degrade sharply under compression and dataset shift.

2.2 Unimodal Audio Spoofing Detection

The anti-spoofing community, organized around the ASVspoof challenges [5], progressed from hand-crafted spectral features to end-to-end models such as RawNet2 [6] and the spectro-temporal graph attention network AASIST [7]. Self-supervised speech front-ends such as WavLM [8] further improved robustness to unseen TTS and VC systems. *Limitation:* audio-only methods cannot detect facial manipulation.

2.3 Audio-Visual Binary Detection

Multimodal detectors fuse both streams but almost universally emit one binary label. The Modality Dissonance Score [9] measures audio-visual dissimilarity; Mittal et

al. [10] compare cross-modal emotional cues; AVFakeNet [11] and AVoiD-DF [12] employ transformer fusion. *Limitation:* no per-modality attribution; a real-video/fake-audio clip and a fake-video/real-audio clip receive the same label.

2.4 Audio-Visual Consistency and Synchronization

SyncNet [13] introduced contrastive audio-visual synchronization, later exploited for anomaly-based deepfake detection (AVAD [14]). *Limitation:* consistency methods detect *mismatch* but cannot say which stream is fake, and they fail on fully-synthetic clips whose fake streams are mutually consistent (e.g., a lip-synced avatar with cloned voice).

2.5 Temporal Localization of Partial Forgeries

LAV-DF [15] introduced content-driven partial forgeries with boundary annotations and the BA-TFD detector; AV-Deepfake1M [16] scaled this to over one million clips. *Limitation:* localization is usually joint over modalities, per-modality attribution is limited, and cross-dataset generalization is under-reported.

2.6 Proxy-Forgery and Generator-Free Training

Self-Blended Images [17] and Face X-ray [18] showed that *synthesizing* blending artifacts from pristine images generalizes better than training on generator outputs; the anti-spoofing literature found the same for vocoder copy-synthesis augmentation [19]. Zhou and Lim [20] explored joint audio-visual labels but trained on generator outputs without a pretraining stage. *Gap:* no prior work unifies the two unimodal proxy-forgery techniques into a quadrant-structured audio-visual pretraining objective, nor introduces a real-but-mismatched class as an explicit disentanglement signal.

2.7 Summary of the Research Gap

Table 2.1 summarizes the identified gap. No widely adopted method simultaneously provides (a) independent per-modality authenticity, (b) cross-modal consistency reasoning, (c) temporal localization, and (d) validated cross-generator generalization in a single reproducible, deployable system. This thesis targets exactly this intersection.

Table 2.1: Capability comparison of prior work and the proposed framework.

Approach	Per-modality	Consistency	Localization	Cross-gen. eval
Unimodal video	–	–	partial	partial
Unimodal audio	–	–	partial	partial
AV binary fusion	–	partial	–	partial
Sync/anomaly based	–	✓	partial	partial
Localization (BA-TFD)	partial	partial	✓	–
DAVID-Net + QACP (ours)	✓	✓	✓	✓

Chapter 3

Proposed Methodology

This chapter presents the proposed unified forensic framework. The framework is trained in two stages: **Stage 0**, Quadrant-Aware Contrastive Pretraining (QACP) on synthetically constructed quadrants; and **Stage 1**, supervised multi-task fine-tuning of the full **DAVID-Net** architecture.

3.1 Problem Formulation

Let a clip $x = (x_v, x_a)$ consist of a visual stream x_v and an acoustic stream x_a . The framework predicts:

$$\hat{y}_v = P(\text{video fake} \mid x) \in [0, 1], \quad (3.1)$$

$$\hat{y}_a = P(\text{audio fake} \mid x) \in [0, 1], \quad (3.2)$$

$$\hat{q} \in \{\text{RVRA}, \text{RVFA}, \text{FVRA}, \text{FVFA}\}, \quad (3.3)$$

$$\hat{m}_v(t), \hat{m}_a(t) \in [0, 1] \quad (\text{per-frame manipulation probability}). \quad (3.4)$$

3.2 DAVID-Net Architecture

3.2.1 Design Principles

DAVID-Net is built on the observation that “is stream X internally authentic?” and “do the two streams agree with each other?” are different questions. Fusing them prematurely allows shortcut learning (e.g., labeling every desynchronized clip fake). The architecture therefore keeps *modality-specific authenticity* evidence disentangled from *cross-modal consistency* evidence, combining them only at the decision heads. This design lets the per-modality heads fire independently—making all four quadrants

reachable, including FVFA clips whose two fake streams are mutually consistent.

3.2.2 Encoders

The video branch extracts aligned face and mouth crops and encodes them with a self-supervised VideoMAE backbone into a spatio-temporal token sequence $V \in \mathbb{R}^{L_v \times d}$. The audio branch encodes the raw 16 kHz waveform with WavLM [8] into $A \in \mathbb{R}^{L_a \times d}$. Encoders are largely frozen; only adapters and the fusion layers are trained heavily.

3.2.3 Synchronization–Consistency Module

Short overlapping windows of V and A are projected into a shared space and trained with an InfoNCE contrastive objective in which temporally aligned real windows are positives. The module outputs a per-window agreement score s_t and a pooled consistency embedding z_c .

3.2.4 Disentangled Cross-Modal Fusion

A cross-modal transformer produces pooled authenticity embeddings z_v, z_a and the consistency embedding z_c . A disentanglement penalty discourages leakage between the authenticity and consistency subspaces:

$$\mathcal{L}_{\text{dis}} = |\cos(z_v, z_c)| + |\cos(z_a, z_c)|. \quad (3.5)$$

3.2.5 Multi-Task Heads and Objective

Four heads consume the disentangled representations: video authenticity $H_v([z_v; z_c])$, audio authenticity $H_a([z_a; z_c])$, quadrant $H_{\text{quad}}([z_v; z_a; z_c])$, and per-frame localization H_{loc} . The total Stage-1 objective is

$$\mathcal{L} = \lambda_1 \mathcal{L}_v + \lambda_2 \mathcal{L}_a + \lambda_3 \mathcal{L}_{\text{quad}} + \lambda_4 \mathcal{L}_{\text{loc}} + \lambda_5 \mathcal{L}_{\text{sync}} + \lambda_6 \mathcal{L}_{\text{dis}}, \quad (3.6)$$

where $\mathcal{L}_v, \mathcal{L}_a$ are focal binary cross-entropy losses, $\mathcal{L}_{\text{quad}}$ is cross-entropy, $\mathcal{L}_{\text{loc}}$ is frame-level binary cross-entropy against manipulated-interval masks, and $\mathcal{L}_{\text{sync}}$ is the contrastive synchronization loss applied to genuine (RVRA) samples.

3.3 Quadrant-Aware Contrastive Pretraining (QACP)

3.3.1 Motivation

Detectors trained on generator outputs inherit those generators’ fingerprints and fail on unseen synthesis methods. QACP instead pretrains the embedding space on forgeries *constructed from pristine data alone*, so the space encodes what manipulation does to a stream rather than what a particular generator looks like.

3.3.2 Synthetic Quadrant Construction

Starting from verified-real (RVRA) clips only, five pseudo-classes are manufactured (Table 3.1).

Table 3.1: QACP pseudo-classes constructed from pristine clips.

Pseudo-class	Construction	Teaches
pseudo-RVRA	untouched real clip	anchor
pseudo-RVFA	vocoder copy-synthesis of the audio	audio authenticity cues
pseudo-FVRA	self-blended face video [17]	video authenticity cues
pseudo-FVFA	both transforms	consistent double-fake
pseudo-MISMATCH	real audio swapped from another clip	mismatch $\neq$ fake

The MISMATCH class is the disentanglement forcing function: both streams remain genuinely real while synchronization is broken, so the authenticity heads must answer “real” while the consistency embedding flags disagreement. Without it, models collapse desynchronization into fakeness—the classic failure mode of synchronization-based detectors.

3.3.3 Factorized Supervised-Contrastive Objective

Three supervised-contrastive (SupCon [21]) losses are applied to the three embedding spaces, each with a *different* label partition of the same batch:

$$\mathcal{L}_{\text{QACP}} = \mathcal{L}_{\text{SupCon}}(z_v; \text{video-auth}) + \mathcal{L}_{\text{SupCon}}(z_a; \text{audio-auth}) + \mathcal{L}_{\text{SupCon}}(z_c; \text{sync-state}). \quad (3.7)$$

Because each space is supervised to be invariant to the other factors, the factorization itself performs disentanglement; the orthogonality penalty of Section 3.2 then only maintains it.

3.4 System Architecture and Deployment

The complete framework is deployed as a two-tier system: a FastAPI inference service hosting a lightweight DAVID-Net variant on Hugging Face Spaces, and a Next.js web interface that presents per-modality verdicts, quadrant attribution, temporal manipulation timelines, and the synchronization curve to end users.

Chapter 4

Experimental Setup

4.1 Datasets

Table 4.1 lists the datasets used. FakeAVCeleb [22] provides native four-quadrant labels and is the primary training set. AV-Deepfake1M [16] and LAV-DF [15] drive the temporal-localization task. DFDC [2] and KoDF are reserved exclusively for cross-dataset evaluation. ASVspoof [5] strengthens the audio branch.

Table 4.1: Datasets used in this study.

Dataset	Modality	Labels	Role
FakeAVCeleb	A+V	4-quadrant	train + in-domain test
AV-Deepfake1M	A+V	clip + localization	train + localization
LAV-DF	A+V	localized forgeries	localization
DFDC	A+V	binary	cross-dataset test only
KoDF	A+V	binary	cross-dataset test only
ASVspoof 2019/21	A	bona fide / spoof	audio-branch auxiliary

4.2 Preprocessing

Video is decoded at 25 fps; faces are detected, tracked, and aligned, producing a 224×224 face crop and a 96×96 mouth crop. Audio is resampled to 16 kHz mono and loudness-normalized. All datasets are converted to a unified manifest schema with per-modality labels, quadrant, manipulated intervals, generator identity, and demographic metadata. Splits are subject-disjoint; leave-one-generator-out (LOGO) splits hold out one manipulation method entirely.

4.3 Implementation Details

The framework is implemented in PyTorch. Experiments run on an NVIDIA DGX Spark system (GB10 Grace-Blackwell, 128 GB unified memory) using BF16 precision. Training follows a cached-feature regime: frozen encoder features are precomputed once, and the fusion module and heads are trained on the cache; the best configuration is optionally fine-tuned end-to-end with LoRA adapters. Optimization uses AdamW (learning rate 10^{-4} for fusion/heads, cosine schedule, 30–50 epochs). Augmentation includes video re-compression, blur, and color jitter, plus audio noise (MUSAN), reverberation, and codec simulation. [TODO: Finalize hyperparameters and GPU-hours after experiments.]

4.4 Evaluation Metrics

- **Detection:** AUC (primary), EER, accuracy, macro-F1 — reported per modality and for the four-class quadrant task.
- **Localization:** AP@IoU {0.5, 0.75, 0.95} and AR@{10, 20, 50}, per modality.
- **Generalization:** cross-dataset AUC and LOGO AUC.
- **Calibration:** Expected Calibration Error and reliability diagrams.
- **Robustness:** AUC versus compression level (H.264 CRF) and audio SNR.
- **Fairness:** AUC/EER gaps across gender, apparent skin tone, and language subgroups.

Statistical rigor: three seeds (mean $\pm$ std), bootstrap 95% confidence intervals on AUC, DeLong tests for AUC differences, and Holm–Bonferroni correction across ablation families.

4.5 Baselines

Unimodal video: Xception [1], EfficientNet-B4, LipForensics [3]. Unimodal audio: RawNet2 [6], AASIST [7]. Multimodal: MDS [9], Emotions Don’t Lie [10], AVoiD-DF [12], and BA-TFD+ [15] for localization. All baselines are evaluated under identical splits and preprocessing.

Chapter 5

Results and Discussion

[TODO: This chapter is populated as experiments complete. The table skeletons below define the paper’s empirical contract.]

5.1 In-Domain Detection and Attribution

Table 5.1: Per-modality and quadrant results on FakeAVCeleb (subject-disjoint test split). [TODO: fill]

Method	Video AUC	Audio AUC	Quadrant Acc.	Macro-F1	EER
Xception (video-only)	–	–	–	–	–
AASIST (audio-only)	–	–	–	–	–
MDS	–	–	–	–	–
AVoiD-DF	–	–	–	–	–
DAVID-Net (ours)	–	–	–	–	–

5.2 Cross-Dataset Generalization

Table 5.2: Cross-dataset AUC: train on FakeAVCeleb + AV-Deepfake1M, test unseen. [TODO: fill]

Method	DFDC	KoDF	DF-TIMIT
Best unimodal baseline	–	–	–
Best multimodal baseline	–	–	–
DAVID-Net w/o QACP	–	–	–
DAVID-Net + QACP (ours)	–	–	–

5.3 Temporal Localization

[TODO: AP@IoU table on AV-Deepfake1M and LAV-DF, per modality, vs. BA-TFD+.]

5.4 Ablation Study

Table 5.3: Ablations. Each row reports in-domain and cross-dataset AUC. [TODO: fill]

Configuration	In-domain AUC	Cross-dataset AUC
Full DAVID-Net + QACP	—	—
– QACP pretraining	—	—
– MISMATCH pseudo-class	—	—
– synchronization module	—	—
– disentanglement loss	—	—
early fusion	—	—
late fusion	—	—
single-task (binary only)	—	—
– localization head	—	—

5.5 Calibration, Robustness, and Fairness

[TODO: Reliability diagrams and ECE; AUC vs. CRF and SNR curves; subgroup gaps.]

5.6 Qualitative Analysis

[TODO: Grad-CAM heatmaps, audio saliency, synchronization curves, localization timelines; include failure cases.]

5.7 Efficiency and Deployment

[TODO: Parameters, FLOPs, latency on DGX Spark (BF16) and the deployed Lite model (FP8/FP4 TensorRT); end-to-end response time of the public web service.]

5.8 Discussion

[TODO: Interpret results against RQ1–RQ6; discuss which hypotheses held; honest treatment of negative results.]

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis presented a unified forensic framework for the detection of multimodal synthetic media that moves beyond binary classification to fine-grained, per-modality attribution: determining independently whether the video and the voice of a clip are real or AI-generated, assigning the clip to one of four authenticity quadrants, and localizing manipulated segments in time. The framework combines DAVID-Net, a disentangled cross-modal transformer, with Quadrant-Aware Contrastive Pretraining, a generator-free pretraining strategy that constructs all four quadrants from pristine data alone. [TODO: Summarize the headline empirical findings once available.] The complete system is released as reproducible open-source code with a deployed public detection service.

6.2 Limitations

- The framework targets single-subject talking-head content; multi-person scenes and non-speech audio are out of scope.
- Detection is probabilistic; outputs are decision support, not proof.
- [TODO: Add empirical limitations discovered during experiments.]

6.3 Future Work

- Extension to fully generated (text-to-video) content without a real source subject.

- Continual adaptation to newly emerging generators using the QACP construction as a self-refreshing training signal.
- Multi-person scenes, overlapping speech, and non-frontal faces.
- Provenance integration (e.g., C2PA content credentials) alongside statistical detection.
- On-device and streaming (real-time) variants of the detector.

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Appendix A

Supplementary Material

A.1 Repository and Reproducibility

The complete source code, configuration files, and dataset split manifests are available at: <https://github.com/MIHMahmudEli/david-net-av>. Every reported number corresponds to a committed configuration file; random seeds are fixed and environments pinned.

A.2 Unified Manifest Schema

Every clip in every dataset is converted to the following record:

```
{
  "clip_id": "str",
  "video_label": 0,           // 0 real, 1 fake
  "audio_label": 1,
  "quadrant": "RVFA",
  "video_segments": [[s, e]], // manipulated intervals (sec)
  "audio_segments": [[s, e]],
  "generator": "wav2lip",
  "dataset": "fakeavceleb",
  "meta": {"gender": "...", "language": "..."}
}
```

A.3 API Response Format

The deployed detection service returns:

```

{
  "video":    {"verdict": "fake", "confidence": 0.94},
  "audio":    {"verdict": "real", "confidence": 0.88},
  "quadrant": {"label": "FVRA", "probs": {...}},
  "localization": {"video": [{"start": 1.2, "end": 3.6,
                             "score": 0.91}], "audio": []},

  "sync_curve": [...],
  "model_version": "david-net-lite-1.0"
}

```

A.4 Additional Results

[TODO: Extended tables, per-generator breakdowns, additional qualitative examples.]